

14.8.4 Maclaurin's Series

when we take $a=0$, then the Taylor's series is called Maclaurin's series. The function $f(x)$ in terms of **Maclaurin's series/ expansion**, under the conditions of Taylor's theorem, is

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \dots + \frac{f^{(n)}(0)}{n!}x^n + \dots$$

... (14.14)

14.8.5 Linear/Quadratic Approximation

Linear approximation

If in (#1) of subsection 14.8.1, we take $k=1$ in $P_{k,a}$, then the *Taylor's polynomial* becomes $P_{1,a} = f(a) + f'(a)(x-a)$ (14.15)

This is called **linear approximation** of the function $f(x)$, provided the values $f(a)$ and $f'(a)$ are Known.

Taylor's formula, for expansion of $f(x)$, in this case, is

$$f(x) = P_{1,a} + R_1(x, a),$$

provided $\lim_{x \rightarrow a} R_1(x, a) = 0$, ... (14.16)

where $R_1(x, a)$ depends on x, a and the function f .

Hence, the **error of approximation** in the computation of $f(x)$ using **linear approximation** is $R_1(x)$.

Quadratic approximation

If in (14.11) of subsection 14.8.1, we take $k=2$ in $P_{k,a}$, then the *Taylor's polynomial*

$$P_{2,a} = f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2$$

... (14.17)

This is called **quadratic approximation** of the function $f(x)$, provided the values $f(a)$, $f'(a)$, and $f''(a)$ are Known.

Taylor's formula, for expansion of $f(x)$, in this case, is

$$f(x) = P_{2,a} + R_2(x, a), \text{ provided } \lim_{x \rightarrow a} R_2(x, a) = 0 \quad \dots (14.18)$$

Hence, the **error of approximation** in the computation of $f(x)$, using quadratic approximation is $R_2(x)$.

Example 14.14

For the function $f(x) = \sin(x)$, find (i) linear and quadratic approximations, and (ii) Maclaurin's series expansion of the function.

Solution:

We calculate values at $x=0$ of $\sin(x)$ and of its various order derivatives, starting with the first derivative.

$$\sin(x)_{x=0} = 0; f'(x)_{x=0} = \cos(x)_{x=0} = 1; f''(x)_{x=0} = -\sin(x)_{x=0} = 0; f'''(x)_{x=0} = -\cos(x)_{x=0} = -1. \quad \dots (14.19)$$

Part (i)

Linear approximation of $f(x) = P_{1,0} = f(0) + x (f'(0))$, where the remainder term $R_1(x)$ is neglected

Linear approximation of $\sin(x) = \sin(0) + x (\cos(x))_{x=0} = 0 + x = x$.

Quadratics approximation of $f(x) = P_{2,0} = f(0) + x f'(x) + (x^2/2!)f''(x)$, where the remainder term $R_2(x)$ is neglected

Quadratics approximation of $\sin x = (\sin x)_{x=0} + x (f'(\sin x))_{x=0} + (x^2/2!)(f''(\sin x))_{x=0}$

$$= 0 + x.1 + ((x^2/2!). 0 = x$$

Part (ii) Maclaurin's expansion

The sequence of values of the derivatives in (14.19) above is repeated for higher order derivatives as follows:

$$f^{iv}(\sin(x))_{x=0} = 0; \quad f^v(\sin(x))_{x=0} = 1; \quad f^vi(\sin(x))_{x=0} = 0; \quad f^{vii}(\sin(x))_{x=0} = -1.$$

The repeating of the above sequence of the values of derivatives may be further generalized as follows:

$$f^{4k}(\sin(x))_{x=0} = 0; \quad f^{4k+1}(\sin(x))_{x=0} = 1; \quad f^{4k+2}(\sin(x))_{x=0} = 0; \quad f^{4k+3}(\sin(x))_{x=0} = -1.$$

for $k = 0, 1, 2, 3, 4, \dots$

Substituting these values in Maclaurin's series expansion given by

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \dots + \frac{f^{(n)}(0)}{n!}x^n + \dots$$

we get

$$\begin{aligned} \sin(x) &= 0 + x.1 + (x^2/2!). 0 + (x^3/3!). (-1) + (x^4/4!). 0 + \dots \\ &= x - (x^3/3!) + (x^5/5!) - (x^7/7!) + (x^9/9!) + \dots \end{aligned}$$

14.9 MULTIVARIATE POLYNOMIAL APPROXIMATION: TAYLOR'S THEOREM, LINEAR APPROXIMATION

In view of the fact that the involved ideas, and even the notation, are quite complex, specially, for the general case, we build up the approximation process gradually, starting with the simplest cases first.

We start our discussion with *polynomial approximation of functions of only TWO variables*. Even this restricted discussion is also carried out in stages. Initially we discuss

14.9.1 Linear Approximation in Terms of Taylor's Polynomials of Order 1

Similar to Taylor's Polynomial of first order, and Taylor's formula, for single variable functions, given by (#3) and (#4) respectively in subsection 14.8.5, **there correspond**

$$P_{1,(a,b)} = f(a,b) + (x-a)f_x(a,b) + (y-b)f_y(a,b), \quad \dots (14.20),$$

the Taylor's **polynomial** of degree ONE, in two variables, x and y, at the point (a, b),

and Taylor's formula for expansion of the function f(x, y) of **two** variables **around (a, b)**, given by

$$\left. \begin{array}{l} f(x, y) = P_{1,(a,b)} + R_1(x, y), \\ \text{provided } \lim_{\substack{x \rightarrow a \\ y \rightarrow b}} R_1(x, y) = 0 \end{array} \right\} \quad \dots (14.21)$$

Thus, **f(x, y)**, is **approximated by polynomial** $P_{1,(a,b)}$,

where

- i) f_x and f_y are the partial derivatives of f(x, y) w. r. t. respectively x and y,
- ii) the values $f(a, b)$, $f_x(a, b)$ and $f_y(a, b)$ at (a, b) of the function f(x, y) and its first order partial **are known**.
- iii) the error in the approximation is $R_1(x, y)$.

The approximation is called **linear** in view of the fact the approximating Taylor's polynomial P_1 is of **degree one**, i. e. is **linear**.

(It may be noted that higher the order of the Taylor's polynomial, lesser is the expected error in the approximation.)

14.9.2 Quadratic Approximation in Terms of Taylor's Polynomials of Order 2, Around (a, b).

Similar to Taylor's Polynomial of **second** order, and Taylor's formula, for **single** variable functions, given by (14.17) and (14.18) respectively in subsection 14.8.5, **there correspond**

$$\begin{aligned} P_{2,(a,b)} = & f(a,b) + \{(x-a)f_x(a,b) + (y-b)f_y(a,b)\} + \{((x-a)^2/2!)f_{xx}(a,b) + (x-a)(y-b)f_{xy}(a,b) + \\ & ((y-b)^2/2!)f_{yy}(a,b)\} \end{aligned} \quad \dots (14.22),$$

Which is the Taylor's polynomial of order 2, **and**

the Taylor's formula of order 2, for expansion of the function f(x, y) of two variables, say x and y, **around (a, b)** given by

$$\left. \begin{array}{l} f(x, y) = P_{2,(a,b)} + R_2(x, y), \\ \text{provided } \lim_{\substack{x \rightarrow a \\ y \rightarrow b}} R_2(x, y) = 0 \end{array} \right\} \quad \dots (14.23)$$

Necessary condition for (14.23) to be true is that f has continuous first-order & second order partial derivatives at (a, b) .

Then $f(x, y)$, is approximated by the Taylor's Polynomial $P_{2, (a, b)}$ given by (14.22)

where

i) f_x and f_y are first order partial derivatives of $f(x, y)$ w. r. t. respectively x

ii) f_{xx} , f_{xy} and f_{yy} are second order partial derivatives of f w. r. t. x and y

iii) the values $f(a, b)$, $f_x(a, b)$, $f_y(a, b)$, $f_{xx}(a, b)$, $f_{xy}(a, b)$, and $f_{yy}(a, b)$, at point (a, b) are known.

iv) the error in the approximation is $R_2(x, y)$.

The approximation is called **quadratic** in view of the fact the approximating Taylor's polynomial P_1 is of **degree two**, i. e. is **quadratic**.

In practical applications, generally we use at most quadratic approximation, however, for theoretical consideration, we discuss here briefly the k^{th} order approximation of a function of n variables, where n stands for 'several'. We state, without proof, the corresponding Taylor's theorem, the proof of which can be obtained in almost straight manner, by generalizing the arguments and conclusions used for linear and quadratic approximations.

Check Your Progress 4

1) Determine all the numbers c which satisfy the conclusions of the Mean Value Theorem for the following function $f(y) = y^3 + 2y^2 - y$ on $[-1, 2]$

.....

2) Verify Mean Value Theorem for the function

$$f(x) = x(x-1)(x-2) \text{ in } [0, 1/2].$$

.....

2) For the function $f(x) = \cos(x)$, find (i) linear and quadratic approximations, and (ii) Maclaurin's series expansion of the function.

.....

- 4) For the function $f(x) = \log x$, around the point $x = 1$, find (i) linear and quadratic approximations, and (ii) Taylor's series expansion of the function.

(Note: $\log(x)$ is NOT defined at $x = 0$. Hence, Maclaurin's series is NOT defined)

.....

14.9.3 General Taylor's Theorem & Approximation of Functions of Several Variables by k^{th} Order Polynomial (optional reading)

Let $F: \mathbf{R}^n \rightarrow \mathbf{R}$, be a function of n variables, $\mathbf{x} = (x_1, \dots, x_i, \dots, x_n)$, and $\mathbf{a} = (a_1, \dots, a_i, \dots, a_n)$, such that $F(\mathbf{x})$, and $F(\mathbf{a}) \in \mathbf{R}$.

Then, *by generalizing to n variables*, and to *k th order Taylor's polynomial*, *the earlier obtained*

- i) the Taylor's polynomial $P_{1, \mathbf{a}}$ and Taylor formula $f(\mathbf{x})$, for *one variable and order k* , given respectively by (14.15) and (14.16), AND
- ii) Taylor polynomial $P_{2, (\mathbf{a}, \mathbf{b})}$ and Taylor formula $f(x, y)$ for 2 variables & order 2 given respectively by (14.17) and (14.18),

we get

$P(\mathbf{x}, \mathbf{a})$

$$= F(\mathbf{a}) + \sum_{i=1} F_{x_i}(\mathbf{a})(x_i - a_i) + \frac{1}{2} \sum_{i,j=1} F_{x_i x_j}(\mathbf{a})(x_i - a_i)(x_j - a_j)$$

$$+ \dots + \frac{1}{k!} \sum_{i_1, \dots, i_k=1}^n F_{x_{i_1} \dots x_{i_k}}(\mathbf{a})(x_{i_1} - a_{i_1}) \dots (x_{i_k} - a_{i_k})$$

... (14.24)

where,

$$F_{x_{i_1} \dots x_{i_k}} = \frac{\partial^k F}{\partial x_{i_1} \dots \partial x_{i_k}}$$

Then general Taylor's theorem for n variables and of order k states that, there exists

$R_k(\mathbf{x}, \mathbf{a})$ such that

$$F(\mathbf{x}, \mathbf{a}) = P(\mathbf{x}, \mathbf{a}) + R_k(\mathbf{x}, \mathbf{a}) \quad \dots (14.25),$$

with $\lim_{\mathbf{x} \rightarrow \mathbf{a}} R_k(\mathbf{x}, \mathbf{a}) = 0$

Hence,

Error of approximation in the computation of $f(\mathbf{x})$, $\mathbf{x} = (x_1, \dots, x_i, \dots, x_n)$, using k th order approximation is $R_k(\mathbf{x}, \mathbf{a})$.

Explanation of the notation in (14.24)

In (14.24), each of the expressions

$$\sum_{i=1}^k F_{x_i}(\mathbf{a})(x_i - a_i)$$

and

$$\frac{1}{2} \sum_{i,j=1}^k F_{x_i x_j}(\mathbf{a})(x_i - a_i)(x_j - a_j)$$

etc. needs explanation.

For the sake of explanation, we take $k=3$.

Then

$$\sum_{i=1}^3 F_{x_i}(\mathbf{a})(x_i - a_i)$$

means

$(\partial F / \partial x_1)(x_1 - a_1) + (\partial F / \partial x_2)(x_2 - a_2) + (\partial F / \partial x_3)(x_3 - a_3)$, Next,

$$\sum_{i,j=1}^3 F_{x_i x_j}(\mathbf{a})(x_i - a_i)(x_j - a_j)$$

means

In the summation, each pair (i, j) takes value $(\partial^2 F / \partial x_i \partial x_j)(x_i - a_i)(x_j - a_j) + (\partial^2 F / \partial x_j \partial x_i)(x_j - a_j)(x_i - a_i)$, and

when $i=j$, then it becomes a single term $(\partial^2 F / \partial x_i^2)(x_i - a_i)^2$.

Similarly,

$$\sum_{i,j,l=1}^k F_{x_i x_j x_l}(\mathbf{a})(x_i - a_i)(x_j - a_j)(x_l - a_l),$$

means

in the summation, each triple (i, j, l) takes value $(\partial^3 F / \partial x_i \partial x_j \partial x_l)(x_i - a_i)(x_j - a_j)(x_l - a_l) + \text{other similar terms obtained by commuting orders of } i, j \text{ and } l$, etc., and in case of any two or three of i, j and l being equal, then similar treatment as above.

And, so on for four or more of suffixes i, j, l, v etc.

The above argument and conclusions may be further generalized to functions $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, where m is any natural number. Generalization is quite

straightforward in the sense that the function $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, with $f(x_1, \dots, x_i, \dots, x_n) = (y_1, \dots, y_i, \dots, y_m)$ can be consider equivalently as

$f = (f_1, \dots, f_j, \dots, f_m)$ with $f_j(x_1, \dots, x_i, \dots, x_n) = y_j$, with $j = 1, 2, \dots, m$. for $(x_1, \dots, x_i, \dots, x_n) \in \mathbb{R}^n$.

Example 14.15: Let $F: \mathbb{R}^3 \rightarrow \mathbb{R}$ such that for $\bar{x} = (x_1, x_2, x_3)$

Let $F(\bar{x}) = e^{x_1 + x_2 + x_3}$. Find Taylor's third order polynomial in the neighbourhood of $(0, 0, 0)$.

For $i=1, 2, 3$, we have $F = e^{x_1 + x_2 + x_3}$;

$$F_{x_i} = \partial F / \partial x_i = e^{x_1 + x_2 + x_3}; F_{x_i x_j} = \partial^2 F / \partial x_i \partial x_j = \partial^2 F / \partial x_j \partial x_i = e^{x_1 + x_2 + x_3};$$

$$F_{x_i x_j x_k} = \partial^3 F / \partial x_i \partial x_j \partial x_k = \dots = \partial^3 F / \partial x_i \partial x_j \partial x_k = e^{x_1 + x_2 + x_3}$$

$$\text{Hence, } (F)_{(0,0,0)} = e^0 = 1, (F_{x_i})_{(0,0,0)} = 1, (F_{x_i x_j})_{(0,0,0)} = 1; (F_{x_i x_j x_k})_{(0,0,0)} = 1, \left. \begin{array}{l} \text{For all } i, j, k \\ \dots (14.26) \end{array} \right\}$$

In (14.26), we substitute $n=3$, $k=3$, $a_1=0$, $a_2=0$, $a_3=0$, $\mathbf{a} = (a_1, a_2, a_3) = (0, 0, 0)$, and values of partial derivatives as given in (14.24) just above, as concluded above, we get

$$\begin{aligned} F(\bar{x}) &= (x_1, x_2, x_3) = F(0, 0, 0) + \sum_{i=1}^3 F_{x_i} (x_i - 0) + (1/2!) \sum_{i=1}^3 \sum_{j=1}^3 F_{x_i x_j} (0, 0, 0) (x_i - 0) (x_j - 0) \\ &+ (1/3!) \sum_{i=1}^3 \sum_{j=1}^3 \sum_{k=1}^3 F_{x_i x_j x_k} (x_i - 0) (x_j - 0) (x_k - 0). \end{aligned}$$

$$= 1 + 1 \cdot (x_1 + x_2 + x_3) + 1/2 (1) \{ (x_1)^2 + x_1 x_2 + x_1 x_3 + x_2 x_1 + (x_2)^2 + x_2 x_3 + x_3 x_1 + x_3 x_2 + (x_3)^2 \}$$

$$+ (1/3!) (1) [(x_1) \{ (x_1)^2 + (x_1)x_2 + (x_1)x_3 + (x_2)^2 + x_2 x_3 + (x_3)^2 \}$$

$$+ (x_2) \{ (x_1)^2 + (x_1)x_2 + (x_1)x_3 + (x_2)^2 + x_2 x_3 + (x_3)^2 \}$$

$$+ (x_3) \{ (x_1)^2 + (x_1)x_2 + (x_1)x_3 + (x_2)^2 + x_2 x_3 + (x_3)^2 \}]$$

$$= 1 + (x_1 + x_2 + x_3) + (1/2!) (x_1 + x_2 + x_3)^2 + (1/3!) (x_1 + x_2 + x_3)^3.$$

14.9.3.1 Taylor's Series Expansion

As in the case of single variables, there are some nice functions of several variables, like e^{x+y+z} which are partially differentiable *infinitely* many times at some points of its domain. In such cases, similar to #2.1 for single variable functions, we can expand such several variable functions, in terms of Taylor's series, around that point by varying k from 1 to ∞ in (\$\$1) as follows

$$\begin{aligned}
&= \mathbf{F}(\mathbf{a}) + \sum_{i=1} \mathbf{F}_{x_i}(\mathbf{a})(x_i - a_i) + \frac{1}{2} \sum_{i,j=1} \mathbf{F}_{x_i x_j}(\mathbf{a})(x_i - a_i)(x_j - a_j) \\
&\quad + \dots + \frac{1}{k!} \sum_{i_1, \dots, i_k=1}^n \mathbf{F}_{x_{i_1} \dots x_{i_k}}(\mathbf{a})(x_{i_1} - a_{i_1}) \cdots (x_{i_k} - a_{i_k}) + \dots
\end{aligned}$$

Example 14.16

Using Taylor's approach, find Taylor's series expansion for the function for

$$f(x, y, z) = xyz,$$

around the point (1, 1, 1). In other words, expansion involves $(x - 1)$, $(y - 1)$, and $(z - 1)$, and partial derivatives of $f(x, y, z)$ at (1, 1, 1)

Let us calculate values of partial derivatives in increasing order:

- $f(1, 1, 1) = 1$.
- First order partial derivatives: $\partial f / \partial x = yz$; $\partial f / \partial y = xz$; $\partial f / \partial z = xy$.

The value of each of these is 1 at (1, 1, 1)

- Second order partial derivatives, for example, $\partial^2 f / \partial x^2 = 0 = \partial^2 f / \partial y^2 = \partial^2 f / \partial z^2$. And $\partial f / \partial y \partial x = z$, $\partial f / \partial z \partial x = y$, and $\partial f / \partial y \partial z = x$, each of which has value 1 at (1, 1, 1).
- Out of third order derivatives, $\partial^3 f / \partial x \partial y \partial z = 1$ and all others are 0.
- All fourth & higher order derivative are 0.

Substituting the values in

$$\begin{aligned}
&= \mathbf{F}(\mathbf{a}) + \sum_{i=1} \mathbf{F}_{x_i}(\mathbf{a})(x_i - a_i) + \frac{1}{2} \sum_{i,j=1} \mathbf{F}_{x_i x_j}(\mathbf{a})(x_i - a_i)(x_j - a_j) \\
&\quad + \dots + \frac{1}{k!} \sum_{i_1, \dots, i_k=1}^n \mathbf{F}_{x_{i_1} \dots x_{i_k}}(\mathbf{a})(x_{i_1} - a_{i_1}) \cdots (x_{i_k} - a_{i_k}) + \dots
\end{aligned}$$

we get

$$\begin{aligned}
f(x, y, z) &= 1 + \{1 \cdot (x - 1) + 1 \cdot (y - 1) + 1 \cdot (z - 1)\} \\
&\quad + (1/2!) \{(x - 1)(y - 1) + (y - 1)(x - 1) + (x - 1)(z - 1) + (z - 1)(x - 1) \\
&\quad + (y - 1)(z - 1) + (y - 1)(z - 1)\} \\
&\quad + (1/3!) \{(x - 1)(y - 1)(z - 1) + (y - 1)(x - 1)(z - 1) + (z - 1)(y - 1)(x - 1)\} \\
&\quad + (1/4!) \cdot 0 + \dots \\
&= 1 + \{(x - 1) + (y - 1) + (z - 1)\} + \{(x - 1)(y - 1) + (x - 1)(z - 1) + (y - 1)(z - 1)\} \\
&\quad + \{(x - 1)(y - 1)(z - 1)\}.
\end{aligned}$$

14.10 LET US SUM UP

In this unit, first of all we clarified the concept of limit, and its significance as the (most) fundamental concept of continuous mathematics. Next, we discussed methods of calculating limits of functions of two or more variables. Then, we introduced the concept of partial derivative and higher order partial derivatives, and discussed methods for calculating these when involved functions are defined (i) explicitly (ii) implicitly and (iii) through functional composition, the last one using chain rule. Next, we defined concepts of directional derivatives, Jacobian Matrix, Hessians, and explained how to use these in solving problems. We discussed Mean Value Theorem, & its applications for single variable and multivariable functions. Finally, we discussed Taylor's theorems, and Taylor's series expansion, both for single variable and multivariable functions, and their applications in approximating functions through polynomial approximation.

14.11 KEY WORDS

Partial derivative

When a variable, say z , is a function of two or more variables, say $x, y, w, \dots$, then a derivative—of the dependent variable z w.r.t any one or more of the variables out of $x, y, w, \dots$, while treating at least one of these as a constant—is called a partial derivative (in our case) of z w. r. t the variables not treated as constants.

Unit vector specification

In n -dimensional space, with $n \geq 2$, when treated as a vector space, with $x_1, \dots, x_i, \dots, x_n$ as independent variables, each treated as an axis, **a unit vector is specified** as $(\cos \theta_1, \cos \theta_2, \dots, \cos \theta_i, \dots, \cos \theta_n)$, where θ_i is the angle which the vector makes with i^{th} axis.

For example, for $n=2$, let X -axis and Y -axis be two mutually perpendicular axes, and let θ be the angle which a unit vector makes with positive direction of X -axis. Then **unit vector's specification** is $(\cos \theta, \cos (\pi/2 - \theta))$.

Gradient of function f of n independent variables, viz. $x_1, \dots, x_i, \dots, x_n$, denoted as $\nabla f(x_1, \dots, x_i, \dots, x_n)$ is the n -tuple $\langle f_{x_1}, \dots, f_{x_i}, \dots, f_{x_n} \rangle$, where f_{x_i} is the partial derivative of f w.r.t. x_i , and $i = 1, 2, \dots, n$.

Directional derivative

In the case of functions of two or more variables, there are *potentially infinite* number of derivatives at a point of the domain: one derivative along one of infinitely many directions. Each such derivative is called a directional derivative.

If unit vector $\vec{u}$ is specified as $\vec{u} = (\cos \theta_1, \dots, \cos \theta_i, \dots, \cos \theta_n)$, and $\nabla f = \langle f_{x_1}, \dots, f_{x_i}, \dots, f_{x_n} \rangle$ is the gradient of f , then **the directional derivative of function f along vector $\vec{u}$** is $\nabla f \cdot \vec{u}$.

Jacobian of the function $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the $m \times n$ matrix of partial derivatives

$$J_f(\bar{x}) = (\partial f_i / \partial x_j)_{i=1,2,\dots,m \text{ and } j=1,2,\dots,n}$$

Hessian: For a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$, which maps $\bar{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ to $f(x) \in \mathbb{R}$

the *Hessian* is, an $n \times n$ square matrix, of all second order partial derivatives of f ,

Mean Value Theorem

For real numbers a and b , with $a < b$, **if** a function $f: [a, b] \rightarrow \mathbb{R}$, is continuous on the closed interval $[a, b]$ and differentiable on the open interval $]a, b[$ **then** there exists a point c in the open interval $]a, b[$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

k^{th} Order Taylor's Polynomial, $P_{k,a}$, involving the available knowledge about values of the derivatives $f^{(j)}(a)$ at the point $x=a$, for $j=0, 1, \dots, k$, as given by

$$P_{k,a} = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^{(k)}(a)}{k!}(x-a)^k.$$

Taylor's formula

The value $f(x)$, **at a general point x** , is given by the following *Taylor's formula*, involving Taylor's Polynomial:

$$f(x) = P_{k,a} + R_{k,a}(x),$$

where, $R_{k,a}(x)$ —which depends on k , the order of Taylor's polynomial, and a , the point at which the values of the function and its derivatives are known—is **called the remainder term**.

14.12 ANSWERS OR HINTS TO CHECK YOUR PROGRESS

Check Your Progress 1

- 1) Determine if the following limit exists or not. If it exists, give the value of the limit.

$$\lim_{(x,y) \rightarrow (4,1)} 3xy^2 / (x+y)$$

Apparently, the function is not continuous along the path $x+y=0$. However, point of our concern, viz. $(4, 1)$ does not lie on it. For the remaining points, including $(4,1)$, in the domain of the function, the function is continuous. Thus, the **limit** of the function is the **value** of the function at $(4,1)$.

$$\lim_{(x,y) \rightarrow (4,1)} 3xy^2 / (x+y) = \lim_{(x,y) \rightarrow (4,1)} 3(4)1^2 / (4+1) = 12/5$$